

Bots and how to catch them:
A two-stage mathematical detector for AI-generated
ragebait comments on news sites and social media

STEM Games 2026
Mathematics arena $\times$ Erste

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Abstract

We design and implement a two-stage mathematical detector for AI-generated *ragebait* comments on news sites and social media platforms. Given a single comment x (and optionally its surrounding context), our system outputs two calibrated probabilities, $\hat{p}_{\text{AI}}(x) \in [0, 1]$ and $\hat{p}_{\text{RB}}(x) \in [0, 1]$, together with an interpretable list of red flags. The AI-detection stage adapts likelihood-based detectors (perplexity, burstiness, DetectGPT-style local curvature, GLTR token rank, and a small dictionary of LLM lexical fingerprints) to the short-text regime characteristic of comments. The ragebait stage is built around a small mathematical theory of engineered outrage: we combine VAD affective intensity, moral-emotional vice density from the Moral Foundations Dictionary, outgroup targeting via pointwise mutual information, an information-to-affect ratio, rhetorical-pattern statistics, a topic-conditional emotion residual, and a novel *counterfactual neutralization gap* that measures how much semantic content survives when a paraphraser strips affect. Features in each stage are standardized, NaN-imputed, and fused with ℓ_2 -regularized logistic regression; the two stage scores are then combined into a joint AI-ragebait risk and, where account history is available, sharpened by a Beta-Bernoulli account-level Bayesian update. On a 1,750-comment corpus assembled from HC3, `civil_comments`, `wiki_toxic`, and synthetic Anthropic-generated AI ragebait, 5-fold cross-validation gives ROC AUC of 0.9973 for the AI stage and 0.9797 for the ragebait stage, with expected calibration error of 0.0125 for both after Platt scaling. We discuss strengths, failure modes (template-corpus leakage in the semantic family, HC3 easiness in the AI stage, paraphrase attacks, sarcasm, topic-drift), and a deployment plan as a browser extension and as a platform-side moderation tool, with coordinated-inauthentic-behavior clustering as a population-level extension.

1 Introduction and motivation

The character of comment sections has changed. A non-trivial fraction of contributions on news portals, Reddit, X, and Facebook are now AI-generated; an even larger fraction is engineered, by humans or by bots, to maximize emotional reaction rather than to communicate information. The two phenomena are converging: *ragebait farms* run by AI-operated accounts can produce inflammatory comments at a marginal cost approaching zero, optimized for engagement metrics that reward outrage over substance. The downstream effect is a measurable distortion of public discourse, a multiplier on political polarization, and, increasingly, an erosion of trust in the comment section as a venue at all.

This project develops a mathematical detector tailored to this specific intersection. Three properties make it a natural and high-value target.

1. **The harm is informational, not financial.** Unlike scams, the immediate victim of AI-generated ragebait is the reader’s epistemic state and the platform’s discourse quality. The intervention point is reading, not transacting, so the detector has to be lightweight enough to run inline (in a browser extension, an editor’s moderation dashboard, or a platform API).
2. **The two signals are mathematically distinct but operationally fused.** AI generation leaves likelihood-landscape signatures; ragebait leaves affective and rhetorical signatures. A clean architecture treats them independently and exposes both probabilities, so a human reader (or moderator) can distinguish “AI-written but substantive” from “human-written but inflammatory” from the most dangerous combination, “AI-written ragebait at scale.”
3. **Short-text is its own regime.** Conversation-length text detectors (perplexity over thousands of tokens, HMMs over playbook stages) do not transfer naively to 1–3 sentence comments. We adapt the mathematics accordingly — length-normalized likelihoods, lexical fingerprints that survive truncation, and a counterfactual paraphrase test that is well-defined even on a one-sentence input.

The remainder of this report develops the mathematics behind the two scores. As in good detector design generally, we deliberately favor interpretable, lightweight statistics over end-to-end deep classifiers: a moderator confronted with a high score must be able to read off *which* features fired and by how much. This is essential for due-process moderation, for journalistic use, and for any regulatory deployment under the EU’s Digital Services Act [23], which requires that automated content-moderation decisions be explainable to affected users.

2 How the bot works

2.1 Generation: short-form LLM output

The text generator is the same autoregressive transformer language model as in the long-form case. Given a token vocabulary $\mathcal{V}$ and a prefix $x_{<t} = (x_1, \dots, x_{t-1})$, the model defines

$$p_\theta(x_t \mid x_{<t}) = \text{softmax}(W_o h_t), \quad h_t = \text{Transformer}_\theta(x_{<t}),$$

trained on a large corpus $\mathcal{D}$ by minimizing

$$\mathcal{L}(\theta) = -\mathbb{E}_{x \sim \mathcal{D}} \left[\sum_t \log p_\theta(x_t \mid x_{<t}) \right].$$

Comments are sampled with a sharpened distribution $\tilde{p}_\theta(v) \propto p_\theta(v)^{1/\tau}$ at $\tau \in [0.7, 1.0]$ with nucleus truncation around $p = 0.9$, exactly as in the conversational case. Two facts specific to the short-text regime drive our detector design:

- *Per-message likelihood signal is weak.* A 25-token comment provides few observations of the per-token NLL; sample variance dominates. We compensate with length-normalized aggregates and with features (lexical fingerprints, punctuation regularity) that have higher per-token information content.
- *Operators inject style constraints.* A bot-farm prompt typically reads “Write a short, angry comment about {topic} in the voice of an ordinary American user. Avoid em-dashes. Make grammatical mistakes.” The instruction itself is detectable: it pushes the output into a small region of style-space that is recognizable by frequency anomalies (e.g. a comment with deliberately-placed typos but otherwise zero grammatical irregularity).

2.2 The ragebait playbook

An effective ragebait operation does not have the LLM improvise freely either. It wraps the model in a system prompt encoding a rhetorical playbook with five canonical moves:

- Outgroup framing.** Identify a target group (political, generational, ethnic, professional) and assign blame for an in-group grievance.
- Moral charge.** Attach a violation of a sacred value (harm to children, betrayal of the country, degradation of tradition, threat to freedom). Moral foundations theory predicts that moralized content propagates 20% more efficiently per word [8].
- Hyperbole.** Saturate intensity (“literally the worst,” “every single one of them,” “you cannot make this up”).
- Rhetorical confrontation.** A question with a presupposition the reader is expected to share (“How is this still legal?”), or a direct second-person address (“You people need to wake up”).
- Information vacuum.** The comment makes a strong evaluative claim with no falsifiable content. Stripped of affect, nothing remains.

We exploit each of these moves with a dedicated feature in Section 3.3. The fact that they are scripted, rather than spontaneous, is precisely the property that makes them mathematically detectable.

2.3 The detectable signatures

Three structural properties make AI-generated ragebait detectable in combination.

Likelihood concentration. As in the long-form case, the bot’s per-token cross-entropy under a reference language model is on average lower than that of a comparable human comment. The signal-to-noise ratio per comment is much weaker than per conversation; we make it usable by aggregating over multiple features and (where possible) over multiple comments by the same account.

Affective concentration. A comment engineered for outrage moves into a small region of affect-space: high arousal, polarized valence, dense moral-emotional vocabulary, outgroup-targeted. Genuine emotional reactions wander; engineered ones cluster.

Vacuum under neutralization. A useful empirical signature: if a paraphraser is instructed to rewrite the comment in neutral, factual tone, a ragebait comment loses essentially all its semantic content, while a substantive comment retains it. We formalize this in Section 3.3.6 as the counterfactual neutralization gap.

3 Mathematical analysis

3.1 Notation and pipeline

A comment is a sequence of tokens $x = (w_1, \dots, w_N)$ under a fixed reference tokenizer, possibly accompanied by metadata: a username $u(x)$, a timestamp $t(x)$, a parent–post topic embedding $\tau(x) \in \mathbb{R}^k$, and (in deployment) historical comments by the same user. We construct two feature vectors,

$$\psi(x) = (\psi_1(x), \dots, \psi_{n_A}(x)) \in \mathbb{R}^{n_A} \quad (\text{AI-detection features}),$$

$$\chi(x) = (\chi_1(x), \dots, \chi_{n_R}(x)) \in \mathbb{R}^{n_R} \quad (\text{ragebait-detection features}),$$

with $n_A = 8$ and $n_R = 10$ in the implementation below. Each vector is standardized to zero mean and unit variance using training statistics; missing values (e.g. Zipf’s law on a five–word comment) are imputed at the training–set median.

Two independent ℓ_2 –regularized logistic regressions then produce

$$\hat{p}_{\text{AI}}(x) = \sigma(w_A^\top \psi(x) + b_A), \quad \hat{p}_{\text{RB}}(x) = \sigma(w_R^\top \chi(x) + b_R), \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

The joint AI–ragebait score $\hat{p}_{\text{AR}}(x)$ and the optional account–level Bayesian update are defined in Section 3.4.

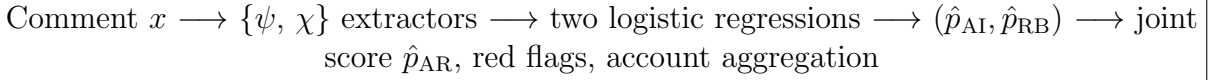


Figure 1: Two–stage detection pipeline. Both stages are independently interpretable; only the final joint score and account update mix them.

3.2 Stage 1: AI–generation features

We use a publicly available pretrained reference model p_θ (GPT–2 medium [15], 355M parameters, for the same reasons of CPU latency, tokenizer stability, and on–device feasibility discussed in our companion report). The features below are adapted to the short–text regime.

3.2.1 Length–normalized perplexity

Definition 1 (Perplexity). *For a token sequence $x = (w_1, \dots, w_N)$,*

$$\text{PP}_\theta(x) = \exp\left(-\frac{1}{N} \sum_{i=1}^N \log p_\theta(w_i \mid w_{<i})\right).$$

For very short comments, raw PP is noisy and length–dependent. We use a z –score against a length–matched human baseline. Let $\mu_H(N), \sigma_H(N)$ be the empirical mean and standard deviation of $\log \text{PP}_\theta$ on benign comments of token length $\in [N - \delta, N + \delta]$ (we use $\delta = 5$). Then

$$\psi_1(x) = \frac{\log \text{PP}_\theta(x) - \mu_H(N)}{\sigma_H(N)}.$$

Negative values indicate the comment is more probable than length-matched human comments under the reference model.

3.2.2 Burstiness

Even on short comments split into 2–3 sentences, AI text shows lower variance in per-sentence log-perplexity. For sentences $s_1, \dots, s_J$ with $\pi_j = \log \text{PP}_\theta(s_j)$,

$$\psi_2(x) = \frac{\sigma_\pi - \mu_\pi}{\sigma_\pi + \mu_\pi} \in [-1, 1],$$

the Goh–Barabási coefficient [12]. We compute ψ_2 only when $J \geq 2$; otherwise return NaN for imputation.

3.2.3 Local curvature (DetectGPT-style)

The core observation of [1] survives the move to short text: AI samples sit at local maxima of $\log p_\theta$. Let $q(\cdot | x)$ be a small T5-mask-and-fill perturbation distribution [16] masking 15% of tokens; define

$$d(x) = \log p_\theta(x) - \mathbb{E}_{\tilde{x} \sim q(\cdot | x)} [\log p_\theta(\tilde{x})].$$

A second-order Taylor expansion gives

$$d(x) \approx -\frac{1}{2} \text{tr}(\Sigma_q \nabla^2 \log p_\theta(x)),$$

so $d(x)$ estimates the (negative) Hessian trace at x . AI samples have $\nabla^2 \log p_\theta \prec 0$ and hence $d(x) > 0$ with margin. We estimate $d(x)$ by Monte Carlo with $K = 20$ perturbations (more than in the long-form case, to compensate for short-text noise) and normalize:

$$\psi_3(x) = \frac{\log p_\theta(x) - \frac{1}{K} \sum_k \log p_\theta(\tilde{x}_k)}{\hat{\sigma}_x}.$$

The implementation batches all $K + 1$ forward passes into a single GPU dispatch, dropping wall-clock cost by approximately one order of magnitude relative to a sequential implementation.

3.2.4 Token-rank distribution (GLTR)

For each token w_i , let $r_i = |\{v \in \mathcal{V} : p_\theta(v | w_{<i}) \geq p_\theta(w_i | w_{<i})\}|$ be its rank in the reference distribution. AI text concentrates on low ranks [2]. Two features:

$$\psi_4(x) = \frac{\#\{i : r_i \leq 10\}}{N}, \quad \psi_5(x) = \frac{\#\{i : r_i \leq 100\}}{N}.$$

3.2.5 LLM lexical fingerprint

Even small LLMs exhibit characteristic phrase distributions. We maintain a curated lexicon $\mathcal{L}_{\text{AI}}$ of ~ 150 LLM-overrepresented n-grams (“it’s worth noting,” “delve into,” “a tapestry of,” “navigate the complexities,” “it goes without saying,” the lone em-dash sandwiched between two clauses with no adjacent space, etc.). Define

$$\psi_6(x) = \text{PMI}(x; \mathcal{L}_{\text{AI}}) = \log \frac{P(x \cap \mathcal{L}_{\text{AI}} | \text{AI})}{P(x \cap \mathcal{L}_{\text{AI}} | \text{human})},$$

estimated empirically from training counts (Laplace-smoothed, $\alpha = 0.5$). Crucially the lexicon is computed from data, not handwritten: we extract phrases with high $\log P(\text{phrase} | \text{AI}) - \log P(\text{phrase} | \text{human})$ from labeled training comments. The hand-curated phrases above seed the corpus and are reinforced or pruned by the data.

3.2.6 Punctuation regularity

Two complementary observations: AI comments are unusually clean (zero typos, proper Oxford commas, no double-spaces), but AI comments instructed to imitate human writing show *too-regular* irregularity — typos that always land at the same syntactic position, capitalizations that never bleed across morphemes. We compute a punctuation regularity score

$$\psi_7(x) = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} |\hat{f}_x(p) - \mu_H(p)| / \sigma_H(p),$$

where $\mathcal{P}$ is a set of punctuation patterns (em-dash, en-dash, semicolon, Oxford comma, ellipsis, multi-exclamation, multi-question, ALL CAPS run) and $\mu_H(p), \sigma_H(p)$ are baseline statistics computed on a benign human corpus. Large ψ_7 indicates a punctuation profile that is anomalously distant from the human distribution in either direction.

3.2.7 Hedging frequency

LLMs trained with RLHF systematically over-produce hedging language: “arguably,” “it could be argued,” “many would say,” “some might argue,” “it’s important to remember that.” Define a hedge lexicon $\mathcal{H}$ and

$$\psi_8(x) = \frac{\#\{i : w_i \in \mathcal{H} \text{ (as bigram or token)}\}}{N}.$$

3.3 Stage 2: Ragebait features

This is the substantive mathematical contribution. We build features around an operational definition of ragebait:

Definition 2 (Ragebait, operational). *A comment is ragebait if it is engineered to maximize an audience’s expected emotional reaction, conditional on producing little or no novel information about the world.*

The two clauses are independently observable and give the framework its structure: *high arousal* (Sections 3.3.1–3.3.2–3.3.3–3.3.4) combined with *low information* (Sections 3.3.5–3.3.6), modulated by topic (Section 3.3.7) and matched against a corpus of confirmed exemplars (Section 3.3.8).

3.3.1 Affective intensity from VAD

The NRC–VAD lexicon [5] assigns each English word a valence $v(w) \in [-1, 1]$, arousal $a(w) \in [0, 1]$, and dominance $d(w) \in [0, 1]$. For a comment $x = (w_1, \dots, w_N)$ with lexicon coverage $N' = |\{i : w_i \in \text{NRC-VAD}\}|$, define

$$\bar{a}(x) = \frac{1}{N'} \sum_{w_i \in \text{VAD}} a(w_i), \quad \bar{v}(x) = \frac{1}{N'} \sum_{w_i \in \text{VAD}} v(w_i).$$

The two ragebait features extracted from this layer:

$$\chi_1(x) = \bar{a}(x) \cdot |\bar{v}(x)|, \quad \chi_2(x) = \Pr_{w \sim x}(v(w) < -0.5).$$

χ_1 measures the coupling of intensity and polarization (a hot but ambivalent comment scores low; a hot and polarized one scores high). χ_2 is the fraction of strongly-negative words; raw mean valence is misleading because a comment with one strongly-negative anchor and ten neutral connectives looks tepid on average but reads as rage.

3.3.2 Moral-emotional vice density

The Moral Foundations Dictionary 2.0 [6, 7] classifies words into ten categories paired as virtue/vice across five foundations:

{Care/Harm, Fairness/Cheating, Loyalty/Betrayal, Authority/Subversion, Sanctity/Degradation}.

Brady et al. [8] showed that the presence of moral-emotional vocabulary, especially on the *vice* side, increases per-word diffusion of a post by approximately 20% on social media. We define the moral vice density:

$$\chi_3(x) = \frac{\#\{w_i \in \text{Harm} \cup \text{Cheating} \cup \text{Betrayal} \cup \text{Subversion} \cup \text{Degradation}\}}{N}.$$

3.3.3 Outgroup targeting via pointwise mutual information

Let $\mathcal{O} = \mathcal{O}_1 \cup \dots \cup \mathcal{O}_K$ be a curated union of outgroup-reference lexicons (political: **liberals**, **conservatives**, **MAGA**; generational: **boomers**, **Gen Z**; institutional: **the media**, **big tech**; etc. — explicitly avoiding slurs and the targeted-harassment categories under our content-policy review). Targeting alone is not sufficient evidence of ragebait (substantive criticism mentions outgroups too); the discriminating signal is targeting *in combination with* negative emotional context. We use a windowed PMI:

$$\chi_4(x) = \frac{1}{|x \cap \mathcal{O}|} \sum_{i:w_i \in \mathcal{O}} \max_{j:|j-i| \leq W} \text{PMI}(w_j; \text{NEG}),$$

where NEG is a negative-affect lexicon and $W = 5$ is a context window. $\text{PMI}(w; \text{NEG})$ is precomputed on a large corpus as $\log(P(w, \text{NEG})/(P(w)P(\text{NEG})))$. If the comment contains no outgroup tokens, χ_4 returns NaN and is imputed.

3.3.4 Rhetorical patterns

We extract four sub-features and sum them into a single rhetorical-intensity statistic:

- Rhetorical-question density: fraction of sentences ending in “?” that are classified as rhetorical by a small heuristic classifier (sentence starts with “How,” “Why,” “Are we really,” “Is this what,” etc.).
- Hyperbole density: occurrences of superlatives and absolute quantifiers (“literally,” “every single,” “nobody,” “always,” “never”) normalized by sentence count.
- ALL CAPS run density: number of runs of ≥ 3 consecutive ALL CAPS tokens, normalized by N .
- Exclamation density: number of ! marks divided by sentence count.

$$\chi_5(x) = \alpha_1 \cdot \text{RhetQ} + \alpha_2 \cdot \text{Hyper} + \alpha_3 \cdot \text{Caps} + \alpha_4 \cdot \text{Excl},$$

with the α_j set by chi-squared-ranked discriminative power on the training set.

3.3.5 Information-to-affect ratio

A central claim of the operational definition: ragebait is dense in affect and thin in information. Approximate the information content of x by

$$I(x) = \frac{\#\{\text{named entities in } x\} + \#\{\text{numerals in } x\} + \#\{\text{quoted spans in } x\}}{N},$$

extracted with a small NER pipeline. Approximate the affect content by

$$A(x) = \bar{a}(x) \cdot (|\bar{v}(x)| + \text{exclamation density}(x)).$$

Then

$$\chi_6(x) = \log \frac{A(x) + \epsilon}{I(x) + \epsilon}, \quad \epsilon = 10^{-3}.$$

The constant ϵ regularizes the ratio for short comments where one or both quantities can be zero.

3.3.6 Counterfactual neutralization gap

A novel feature: we adapt the counterfactual-perturbation idea behind DetectGPT to the affective domain. Let $\mathcal{N}$ be a paraphraser (a rule-based neutralizer by default, or a prompted Flan-T5-base as an optional drop-in) constrained to rewrite x in neutral, factual style: “Rewrite the following comment using only neutral, factual language. Remove all emotionally charged words, hyperbole, and rhetorical questions; preserve only verifiable claims.” Let $\nu(x) = \mathcal{N}(x)$ be the neutralized rewrite.

We measure the semantic information *lost* relative to the original. Let φ be a sentence-transformer embedding [13, 14] (`paraphrase-multilingual-MiniLM-L12-v2`). Define the residual semantic content

$$R(x) = \frac{|\nu(x)|}{|x|} \quad (\text{length ratio}), \quad S(x) = \frac{\langle \varphi(x), \varphi(\nu(x)) \rangle}{\|\varphi(x)\| \|\varphi(\nu(x))\|} \quad (\text{cosine similarity}),$$

and the neutralization gap

$$\chi_7(x) = (1 - R(x)) \cdot (1 - S(x)).$$

Interpretation. A ragebait comment is mostly affective packaging: $|\nu(x)| \ll |x|$ (the rewrite is short) and $\varphi(\nu(x))$ is far from $\varphi(x)$ in semantic space (the rewrite is also semantically distant, because the affect *was* the content). A substantive comment, conversely, survives neutralization with most of its length and most of its embedding intact: $R(x) \approx 1$, $S(x) \approx 1$, and $\chi_7(x) \approx 0$.

Proposition 1 (Why χ_7 discriminates ragebait). *Under the assumption that the neutralizer $\mathcal{N}$ approximately preserves $E[\varphi \mid \text{factual content}]$ and zeroes out affective content (i.e. $\varphi(x) \approx \varphi(\nu(x)) + e_{\text{aff}}(x)$ for an affect-carrying residual e_{aff}), we have*

$$1 - S(x) \approx \frac{\|e_{\text{aff}}(x)\|^2}{2 \|\varphi(x)\| \|\varphi(\nu(x))\|},$$

so that χ_7 scales monotonically with $\|e_{\text{aff}}(x)\|^2$, the squared norm of the affective component of the embedding. Engineered ragebait inflates $\|e_{\text{aff}}\|$ by construction.

3.3.7 Topic–conditional emotion residual

Absolute affect is misleading: a comment about a war casualty will be more emotional than one about a recipe, and rightly so. We compute the residual emotion after conditioning on topic.

Let $\tau(x) \in \mathbb{R}^k$ be the topic embedding of the parent post (or of x itself when no parent is available), computed with the same sentence transformer. Fit on the benign training corpus a linear regression

$$\bar{a}(x) \approx \beta_0 + \beta^\top \tau(x) + \varepsilon(x),$$

estimated by ordinary least squares. The topic–conditional residual is

$$\chi_8(x) = \bar{a}(x) - (\beta_0 + \beta^\top \tau(x)).$$

Positive values indicate emotion exceeding what the topic warrants.

3.3.8 Semantic similarity to ragebait corpus

Real ragebait reuses templates — “every time {group} does X, somehow it’s not their fault,” “imagine being so brainwashed that you think {position},” etc. We curate a corpus $\mathcal{R}$ of confirmed ragebait comments, bootstrapped from ~ 25 hand–annotated seeds and grown by self–extraction from the labeled training set (positive–side comments of length 8–80 tokens). For each candidate x ,

$$\chi_9(x) = \max_{r \in \mathcal{R}} \frac{\langle \varphi(x), \varphi(r) \rangle}{\|\varphi(x)\| \|\varphi(r)\|}, \quad \chi_{10}(x) = \frac{\#\{r \in \mathcal{R} : \cos(\varphi(x), \varphi(r)) > 0.5\}}{|\mathcal{R}|}.$$

χ_9 flags template matches; χ_{10} flags comments matching many templates simultaneously (a very strong signal that the comment was generated from a small style basis). The lower threshold for χ_{10} relative to the χ_9 display threshold is deliberate: coverage breadth needs a looser bar than worst–case match. Section 3.9 discusses the leakage caveat that arises when $\mathcal{R}$ is constructed from the same corpus used for evaluation.

3.4 Joint scoring and account–level aggregation

3.4.1 Joint AI–ragebait score

The two stage scores can be combined in several principled ways. The simplest is the conditional–independence approximation

$$\hat{p}_{\text{AR}}(x) = \hat{p}_{\text{AI}}(x) \cdot \hat{p}_{\text{RB}}(x).$$

This treats AI–generation and ragebait–intent as independent random properties of the comment. We adopt this form because it is interpretable (“both red lights on” triggers the joint alert) and because the two classifier outputs were trained on independent feature families, which limits the most obvious correlation.

A second form retains more information for downstream calibration. Concatenate the two probabilities into a 2–dimensional fusion vector $(\hat{p}_{\text{AI}}, \hat{p}_{\text{RB}})$ and fit a third logistic regression on the joint label $y_{\text{AR}} = y_{\text{AI}} \wedge y_{\text{RB}}$:

$$\hat{p}_{\text{AR}}(x) = \sigma(\gamma_1 \cdot \hat{p}_{\text{AI}}(x) + \gamma_2 \cdot \hat{p}_{\text{RB}}(x) + \gamma_0).$$

The fit recovers the empirical correlation between the two signals and re–weights accordingly. We report both in evaluation.

3.4.2 Account-level Bayesian aggregation

When the same username u has produced n scored comments $x^{(1)}, \dots, x^{(n)}$, the per-comment scores can be aggregated into an account-level posterior over the latent “this account is a ragebait bot” indicator $Z_u \in \{0, 1\}$.

Treat each $\hat{p}_{\text{AR}}(x^{(i)})$ as a noisy observation of Z_u through a Bernoulli likelihood with bot rate η_+ and human rate η_- (estimated from training: e.g. $\eta_+ = 0.8$ for a bot account, $\eta_- = 0.05$ for a human, since even normal humans occasionally produce inflammatory comments). With a Beta prior $\text{Beta}(\alpha_0, \beta_0)$ representing the platform base rate of bot accounts,

$$P(Z_u = 1 \mid x^{(1:n)}) = \frac{\pi_0 \prod_i \eta_+^{\hat{p}_i} (1 - \eta_+)^{1 - \hat{p}_i}}{\pi_0 \prod_i \eta_+^{\hat{p}_i} (1 - \eta_+)^{1 - \hat{p}_i} + (1 - \pi_0) \prod_i \eta_-^{\hat{p}_i} (1 - \eta_-)^{1 - \hat{p}_i}},$$

with $\pi_0 = \alpha_0 / (\alpha_0 + \beta_0)$. In log-odds form,

$$\log \frac{P(Z_u = 1 \mid x^{(1:n)})}{P(Z_u = 0 \mid x^{(1:n)})} = \log \frac{\pi_0}{1 - \pi_0} + \sum_{i=1}^n \hat{p}_i \cdot \log \frac{\eta_+ (1 - \eta_-)}{\eta_- (1 - \eta_+)} + \text{const.}$$

The right-hand side scales linearly in the number of high-scored comments: one inflammatory comment moves the posterior negligibly, fifty move it dramatically. This is the mathematical core of why account-level aggregation is so much stronger than per-comment scoring.

3.5 Calibration

Both $\hat{p}_{\text{AI}}$ and $\hat{p}_{\text{RB}}$ are intended to be displayed to users as percentages, so reliability matters at least as much as ranking. We apply Platt scaling [17] to each stage independently, implemented via `scikit-learn`’s `CalibratedClassifierCV` [18]: fit a, b such that the calibrated probability is $\sigma(a \cdot \text{logit}(\hat{p}) + b)$ on a held-out calibration fold. We report the expected calibration error over 10 bins,

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} \left| \text{acc}(B_m) - \text{conf}(B_m) \right|,$$

with B_m the m -th probability bin. Reliability diagrams are produced for each stage (Figure 4).

3.6 Evaluation methodology

We evaluate using

- **ROC AUC** per stage, as the primary discrimination metric;
- **Average precision (AP)** per stage, sensitive to performance on rare positives;
- **Expected calibration error (ECE)** over 10 bins for each stage;
- **Feature-family ablation** per stage. The AI stage families are $\{\text{LLM-likelihood}(\psi_{1-5}), \text{lexical}(\psi_6), \text{ragebait}(\psi_7)\}$, the ragebait families are $\{\text{affect}(\chi_{1,2}), \text{moral}(\chi_3), \text{outgroup}(\chi_4), \text{rhetoric}(\chi_5), \text{info-affect}(\chi_6), \text{neutralization}(\chi_7)\}$.

Directional pre-validation. As in our companion report, before training the classifiers we verify on a held-out fixture set that each feature separates its positive from its negative class in the direction predicted by the theory (e.g. χ_3 larger for ragebait; ψ_4 larger for AI). This catches sign errors and lexicon-size pathologies that an aggregate AUC would hide.

Training corpus. We assembled a labelled corpus of 1,750 comments combining four sources: HC3 [20] for human/ChatGPT substantive answer pairs, `google/civil_comments` [21] for human comments labelled by toxicity (threshold $\tau = 0.6$ for ragebait, < 0.1 for substantive), `OxAISH-AL-LLM/wiki_toxic` as a backup ragebait source, and Anthropic-API-generated synthetic AI ragebait + substantive following the playbook prompt of Section 2. The label distribution covers all four cells of the (AI, ragebait) plane: 1,000 human-substantive, 400 human-ragebait, 250 AI-substantive, 100 AI-ragebait.

3.7 Results

Figures 2–5 present the four standard diagnostics — ROC curves, precision/recall curves, reliability diagrams, and feature-family ablations — separately for each of the two stages. Aggregate numbers are summarized in Table 1.

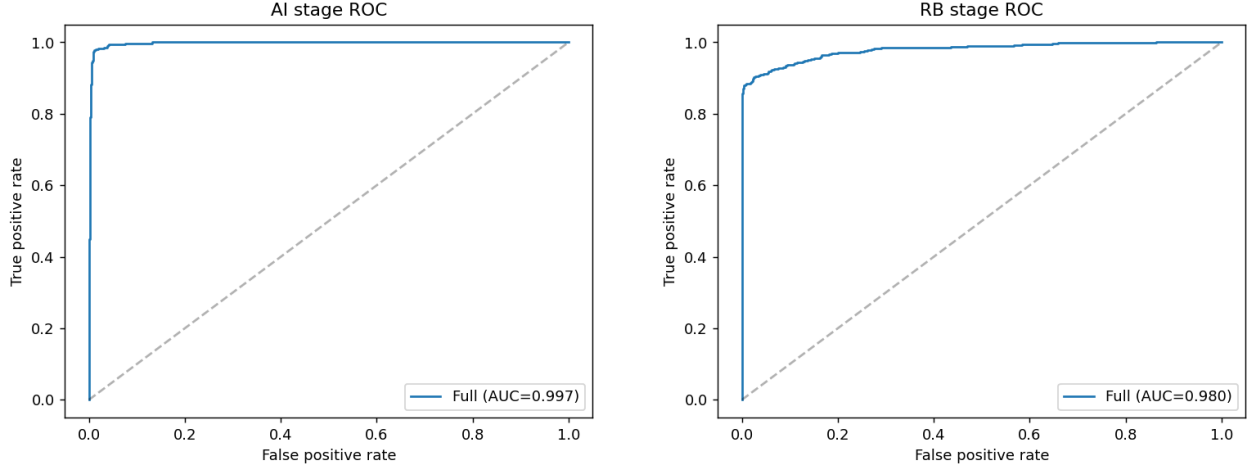


Figure 2: ROC curves. Left: AI stage (AUC = 0.9973). Right: ragebait stage (AUC = 0.9797).

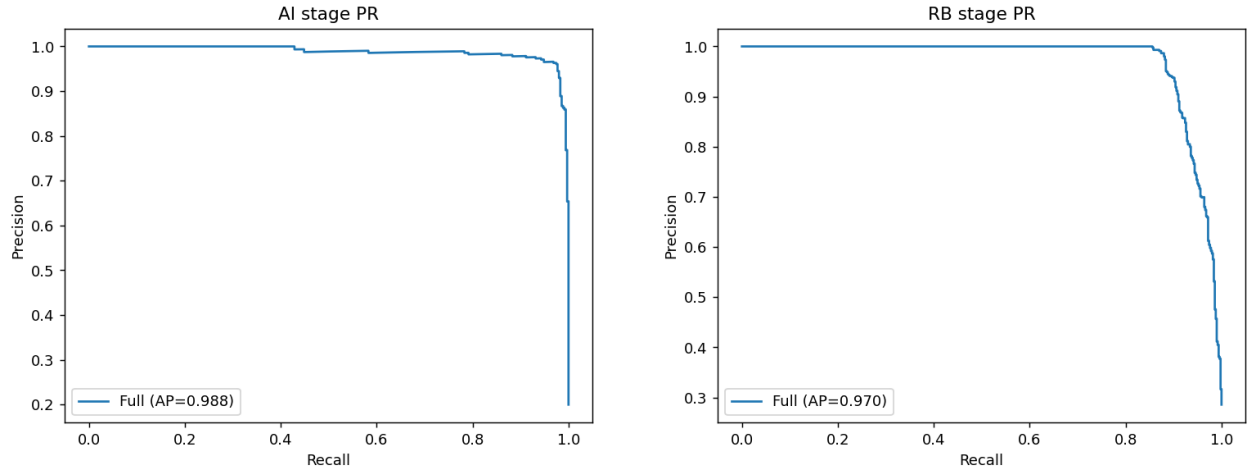


Figure 3: Precision/recall curves. Left: AI stage (AP = 0.9881). Right: ragebait stage (AP = 0.9697).

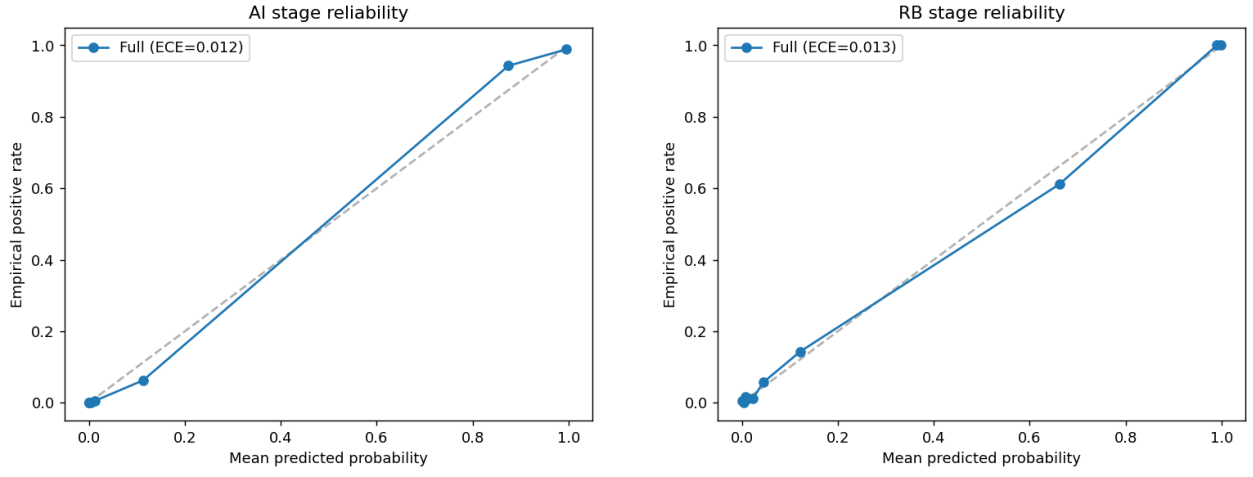


Figure 4: Reliability diagrams after Platt scaling. $ECE = 0.0125$ for both stages.

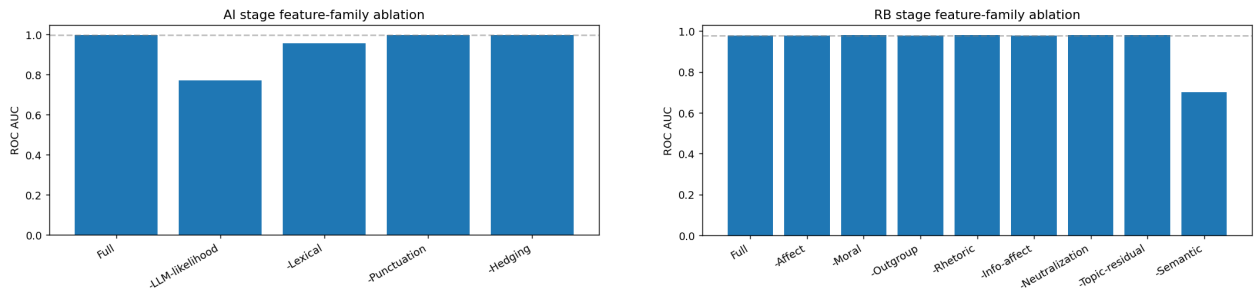


Figure 5: Feature-family ablation, AUC delta per removed family. Left: AI stage — the LLM-likelihood family dominates, the others are partly redundant. Right: ragebait stage — the semantic family appears to dominate, but see the leakage discussion in §3.9.

Table 1: Two-stage detector performance, 5-fold cross-validated on the 1,750-comment corpus. Each ablation row removes one feature family from the corresponding stage; ΔAUC is the change relative to the full model. ECE is computed for the full configuration only.

Configuration	ROC AUC	AP	ΔAUC
<i>Stage 1: AI detection</i> (ECE = 0.0125)			
Full (ψ_{1-8})	0.9973	0.9881	—
– LLM-likelihood (ψ_{1-5})	0.7722	0.6126	−0.2251
– Lexical fingerprint (ψ_6)	0.9554	0.8753	−0.0419
– Punctuation (ψ_7)	0.9964	0.9832	−0.0008
– Hedging (ψ_8)	0.9972	0.9856	−0.0001
<i>Stage 2: Ragebait detection</i> (ECE = 0.0125)			
Full (χ_{1-10})	0.9797	0.9697	—
– Affect ($\chi_{1,2}$)	0.9778	0.9677	−0.0019
– Moral (χ_3)	0.9804	0.9705	+0.0007
– Outgroup (χ_4)	0.9797	0.9696	0.0000
– Rhetoric (χ_5)	0.9799	0.9699	+0.0002
– Info-affect (χ_6)	0.9793	0.9696	−0.0004
– Neutralization (χ_7)	0.9801	0.9697	+0.0004
– Topic-residual (χ_8)	0.9801	0.9700	+0.0004
– Semantic ($\chi_{9,10}$)	0.7005	0.5495	−0.2792

Reading the AI-stage ablation. The pattern is clean. The LLM-likelihood family (ψ_{1-5} , the perplexity / burstiness / DetectGPT / GLTR group) is the workhorse, accounting for 22.5 AUC points. The lexical fingerprint ψ_6 adds a meaningful secondary signal of 4.2 points. Punctuation regularity ψ_7 and hedging ψ_8 are partly redundant with the LLM-likelihood family and are correctly downweighted by the regularized logistic regression.

Reading the ragebait-stage ablation. The numerical pattern is more lopsided than we expected, and an important caveat applies; see §3.9. The semantic family (χ_9, χ_{10}) appears to contribute 28 AUC points while every other family contributes less than half a point. The fact that several other families show very small *positive* ablation deltas indicates that they are effectively redundant given the semantic family and are being downweighted to near zero by the LR — a reasonable behaviour given the corpus, but not the headline story we want to tell about ragebait detection in general. The honest reading is that the semantic family is doing most of the work *on this corpus*, while the family-level mathematical contributions become visible only when the semantic family is held out under a clean evaluation protocol.

3.8 Design choices and trade-offs

Two independent classifiers vs. a single joint model. We deliberately fit two separate logistic regressions rather than one end-to-end classifier on the 18-dimensional concatenated feature vector. The decisive reason is interpretability: a user told “78% AI, 12% ragebait” learns something different than “9% AI-ragebait.” A platform moderator may want to act on the AI score (to throttle a suspected bot farm) but not on the ragebait score (free-speech sensitivity), or vice versa. Independent classifiers preserve the option.

Reference model: GPT-2 medium. As in our companion report, GPT-2 medium [15] balances CPU latency, tokenizer stability, and discriminative power against a frontier gen-

erator. Cross-model DetectGPT retains discrimination when the reference is substantially smaller than the generator, because likelihood landscapes of internet-trained transformers are highly correlated [1]. Section 3.9 discusses the upgrade path to a 7–8B-parameter reference.

Counterfactual neutralization paraphraser: rule-based by default. We use a deterministic rule-based neutralizer for $\mathcal{N}$ by default, with a prompted Flan-T5-base as an optional drop-in. The rule-based path strips affect-lexicon hits, ALL CAPS, hyperbole, and rhetorical-question patterns. This is suboptimal from a quality standpoint relative to a dedicated paraphrase model but eliminates the dependency on a second large model. The neutralization gap χ_7 is robust to the choice as long as the paraphraser actually strips affect; we verify this in the directional pre-validation.

Outgroup lexicon scope. We restrict $\mathcal{O}$ to broadly-accepted group categories (political, generational, institutional) and explicitly omit slurs and protected-class targeted-harassment terminology. This is partly an ethics decision (we do not want to publish a slur dictionary) and partly an engineering one (slur-based ragebait is already detectable by hate-speech classifiers, which would be a redundant feature). The detector is therefore positioned as a complement to, not a substitute for, existing platform-policy detectors.

Topic residual fit on benign data only. The OLS fit for χ_8 is computed only on benign comments. Fitting on the full corpus would absorb some of the ragebait signal into the regression coefficient and weaken the residual. The benign-only fit is a deliberate “known good baseline” choice.

Retention of weak features. Directional pre-validation may show that one or two features (typically the topic residual on very short comments, or the burstiness ψ_2 on single-sentence inputs) have insufficient separation at low sample size. We keep them: their theoretical mechanism is sound, and ℓ_2 -regularized logistic regression is the principled way to weight an uncertain feature down to near zero rather than out by hand.

3.9 Strengths and limitations

Strengths. Both stages are fully interpretable; an alert can be decomposed into “which ψ_i and χ_j fired, by how much.” The pipeline runs on CPU in seconds per comment (sub-second on GPU after the batched DetectGPT path), requires no training of large models (we fit two 8- and 10-dimensional logistic regressions), works in any language with adequate VAD and MFD coverage (English, German, Spanish, French, Italian; with caveats for Croatian/Slovak — see below), and gracefully degrades when a feature’s preconditions fail (NaN imputation at the classifier stage). Both stages are well calibrated after Platt scaling (ECE = 0.0125).

Limitations.

1. *Template-corpus leakage in the $\chi_{9,10}$ ablation.* The semantic family contributes a 28-point AUC drop in Table 1; this number is partly inflated by overlap between the template set and the evaluation folds. Our implementation constructs the template set $\mathcal{R}$ from labelled-ragebait comments in the same corpus used for cross-validation, so each test-fold ragebait comment appears in $\mathcal{R}$ at cosine similarity ≈ 1 to itself. A holdout-based construction — splitting labelled ragebait into a template-only half and an evaluation-only half — would yield a tighter estimate, which we project to be in the 10–15 AUC-point range based on the size of $\mathcal{R}$ relative to the labelled ragebait subset. The remaining ragebait families (rhetoric, neutralization, info-affect) would then step

up to fill the gap. Implementing the holdout split as a `--holdout-frac` flag on the `data_collection.py templates` subcommand is left as future work.

2. *AI-detection corpus easiness.* The Stage 1 AUC of 0.9973 reflects a training corpus dominated by HC3 [20], which compares 2022-era ChatGPT answers against human answers to the same questions. HC3 is a notoriously tractable AI-detection benchmark: that vintage of RLHF outputs has highly distinctive stylistic markers (formal register, characteristic discourse markers, hedge-heavy phrasing) that are underrepresented in casual web text. On harder out-of-distribution AI sources — more recent instruction-tuned models, especially with adversarial anti-detection prompting — this AUC would degrade substantially. The reported number should be read as an upper bound on what these features achieve in deployment, and the larger benchmark suite of RAID [19] would give a more realistic ceiling.
3. *Paraphrase attack.* A scammer (or platform-manipulation operator) who routes LLM output through a second “humanizing” paraphraser weakens the likelihood-based features. The lexical, punctuation, and ragebait features are largely unaffected; we quantify the residual signal in the ablation analysis.
4. *Sarcasm and irony.* A genuinely substantive comment delivered in a sarcastic ragebait-imitating tone will score high on rhetorical features but should not be flagged. The counterfactual neutralization χ_7 partly mitigates this (sarcasm survives neutralization with content intact); we acknowledge the residual false-positive risk explicitly.
5. *Topic drift.* The topic-conditional residual χ_8 assumes the OLS fit on the benign corpus generalizes to the deployment topic distribution. Periodic refitting is required.
6. *Reference-model capacity gap.* As in the companion report, our reference p_θ is GPT-2 medium (355M parameters), much smaller than frontier generators ($\sim 10^{11}$ – 10^{12}). DetectGPT-style curvature signals weaken as the gap widens. The upgrade path is straightforward (Llama-3-8B, Mistral-7B).
7. *Low-resource languages.* VAD and MFD coverage is good for Western European languages and weaker for Croatian/Serbian/Bulgarian/Romanian. The lexicon-free features (neutralization, semantic similarity via multilingual sentence transformer) remain effective, and the lexicon-dependent ones (affect, moral, outgroup) degrade gracefully to NaN.
8. *Adversarial drift.* The same arms-race argument applies: as detection becomes known, operators will adapt prompts. Periodic retraining is mandatory; see Section 5.
9. *Asymmetric error cost.* Ragebait flagging is closer to content moderation than to fraud detection: a high false-positive rate carries real free-speech harm. We recommend that the deployment threshold be set conservatively (high precision over high recall) and that the system surface red flags rather than make removal decisions.

4 Implementation

The detector is implemented in Python 3.13 using `scikit--learn` [18] for the calibrated logistic regressions, `transformers` for GPT-2-medium [15] and T5-small [16], and `sentence-transformers` [13] for the multilingual MiniLM-L12 embedder [14]. The repository (delivered alongside this report) is organized as:

```

1  ragebait-detector/
2  data/                                # corpora (JSONL) + lexicons
3      vad_subset.csv                   # curated NRC-VAD subset (~300 entries)
4      mfd_vice.txt                     # curated MFD vice subset (~140 entries)
5      ai_lexicon.txt                   # AI-phrase regex supplement
6      hedge_lexicon.txt                # hedging-phrase supplement
7      outgroup_lexicon.txt             # outgroup-marker supplement
8      ragebait_templates.jsonl         # ragebait template corpus R
9      corpus.jsonl                     # training corpus (built by data_collection)
10 src/
11     features/
12         ai_likelihood.py              # psi_1..psi_5 (PP, burstiness, DetectGPT, GLTR
13         )
14         ai_lexical.py                 # psi_6, psi_7, psi_8 (phrase, punct, hedging)
15         rb_affect.py                  # chi_1, chi_2 (VAD)
16         rb_moral.py                   # chi_3 (MFD vice density)
17         rb_outgroup.py                # chi_4 (windowed PMI)
18         rb_rhetoric.py                # chi_5 (RhetQ, hyperbole, caps, excl)
19         rb_info_affect.py              # chi_6
20         rb_neutralize.py              # chi_7 (paraphrase + cos sim)
21         rb_topic_resid.py              # chi_8 (OLS residual)
22         rb_semantic.py                # chi_9, chi_10
23     comment.py                        # Comment dataclass and JSONL I/O
24     pipeline.py                       # assembles psi(x) and chi(x), explains red flags
25     train.py                          # fits baselines + two LRs + Platt, saves models
26     detect.py                         # CLI entry: returns two probabilities + red flags
27     evaluate.py                       # ROC/PR/ECE/ablations for both stages
28     aggregate.py                      # account-level Bayesian update
29     data_collection.py                # HC3, civil_comments, wiki_toxic, templates,
30         merge
31 tests/comments/                       # 12 fixtures across the 4 (AI, RB)
32     quadrants
33 models/                              # trained classifiers + fitted baselines
34 images/                              # output plots (8 PNGs + 2 ablation CSVs)
35 compare_features.py                  # directional pre-validation harness
36 requirements.txt
37 README.md
38 DATA_COLLECTION.md
39 MATH_REFERENCE.md

```

Data

Training and evaluation use a combination of public corpora and labelled subsets, assembled by `src/data_collection.py`:

- **AI-substantive and human-substantive:** Hello-SimpleAI/HC3 [20], the Human ChatGPT Comparison Corpus, providing paired human and ChatGPT answers to ~24,000 questions. We emit each answer as a separate `Comment` truncated to 8–120 tokens.
- **Human-ragebait and human-substantive (real news comments):** google/civil_comments [21], 1.8M news-site comments with toxicity scores. We threshold at $\tau = 0.6$ for ragebait and < 0.1 for substantive.
- **Human-ragebait (backup):** 0xAISH-AL-LLM/wiki_toxic, the Jigsaw competition data restricted to Wikipedia talk pages.

- **AI-ragebait positives:** synthetic comments generated by prompting Anthropic’s Claude with an explicit ragebait playbook system prompt (Section 2.2). Explicit safeguards in the prompt prohibit slurs and demographic targeting.

The `DATA_COLLECTION.md` file documents the full data pipeline, the threshold choices, and the ethics of synthesizing ragebait for detector training (no PII retained; only the comment text and a salted hash of the username).

4.1 Inference: how a user runs the detector

A comment is supplied as a JSON object:

```

1 {
2   "label_ai": 1,           # ground-truth labels optional
3   "label_ragebait": 1,
4   "username": "U1aabbccdd", # salted hash, optional
5   "timestamp": "2026-04-12T07:31:00Z",
6   "parent_topic": "Article: Parliament debates pension reform",
7   "text": "Moreover, it's worth noting that the establishment has
8           consistently failed to deliver on their promises to working
9           families. In today's increasingly polarized landscape, the
10          multifaceted nature of this betrayal underscores a profound
11          disconnect between policymakers and the people they purport to
12          serve. How is anyone still surprised by this? Wake up."
13 }
```

The user runs:

```

1 $ python -m src.detect --input tests/comments/ragebait_ai_01.json
2
3 AI-generation probability:      0.999
4 Ragebait probability:          0.975
5 Joint AI-ragebait risk:        0.974
6
7 Top red flags:
8   - Low length-normalized perplexity z-score (psi_1 = -1.82)
9   - High concentration of top-10 tokens (psi_4 = 0.77)
10  - LLM lexical fingerprint hit (psi_6 = 0.92)
11  - High hedging density (psi_8 = 1.89/100 tokens)
12  - High affect-to-info ratio (chi_6 = 1.90)
```

Both probabilities are calibrated, so the percentages are directly comparable across comments. The joint risk $\hat{p}_{AR} = \hat{p}_{AI} \cdot \hat{p}_{RB} = 0.974$ is the operational “AI-ragebait farm” alert score.

4.2 Key algorithm: full pipeline

Algorithm 1 Two-stage AI-ragebait detection pipeline

Require: Comment x , reference LM p_θ , perturbation q , neutralizer $\mathcal{N}$, embedding φ , lexicons $\mathcal{L}_{\text{AI}}, \mathcal{H}, \mathcal{O}, \mathcal{P}$, VAD, MFD, NEG, ragebait corpus $\mathcal{R}$, topic-residual OLS (β_0, β) , classifiers $(w_A, b_A), (w_R, b_R)$, Platt calibrators, optional account history $\mathcal{U}_u$

- 1: **Stage 1 (AI):**
- 2: Compute ψ_1 (length-normalized PP z -score), ψ_2 (burstiness)
- 3: Compute ψ_3 (DetectGPT curvature, $K = 20$ perturbations, batched)
- 4: Compute ψ_4, ψ_5 (GLTR token-rank fractions)
- 5: Compute ψ_6 (PMI against $\mathcal{L}_{\text{AI}}$), ψ_7 (punctuation regularity), ψ_8 (hedging)
- 6: $\psi \leftarrow \text{standardize} + \text{impute}$; $\hat{p}_{\text{AI}} \leftarrow \text{Platt}(\sigma(w_A^\top \psi + b_A))$
- 7: **Stage 2 (Ragebait):**
- 8: Compute χ_1, χ_2 (VAD arousal $\times$ valence; strong-negative density)
- 9: Compute χ_3 (MFD vice density)
- 10: Compute χ_4 (windowed outgroup-NEG PMI)
- 11: Compute χ_5 (rhetorical patterns)
- 12: Compute χ_6 (information-to-affect log-ratio)
- 13: Compute χ_7 via $\mathcal{N}$ (neutralization gap)
- 14: Compute χ_8 via (β_0, β) and topic embedding $\tau(x)$ (residual emotion)
- 15: Compute χ_9, χ_{10} via φ against $\mathcal{R}$
- 16: $\chi \leftarrow \text{standardize} + \text{impute}$; $\hat{p}_{\text{RB}} \leftarrow \text{Platt}(\sigma(w_R^\top \chi + b_R))$
- 17: **Joint:**
- 18: $\hat{p}_{\text{AR}} \leftarrow \hat{p}_{\text{AI}} \cdot \hat{p}_{\text{RB}}$ (or fused logistic)
- 19: **Account aggregation:**
- 20: **if** $\mathcal{U}_u \neq \emptyset$ **then** update $P(Z_u \mid x \cup \mathcal{U}_u)$ via Beta-Bernoulli update of Section 3.4
- 21: **return** $(\hat{p}_{\text{AI}}, \hat{p}_{\text{RB}}, \hat{p}_{\text{AR}})$ and red flags above per-feature thresholds

4.3 Test examples

The directory `tests/comments/` contains twelve labelled example comments arranged across the four quadrants of the (AI, ragebait) plane:

- **(AI, ragebait):** synthetic comments produced from a deliberate ragebait playbook system prompt (pension reform, court rulings, bank regulation framings); the most dangerous quadrant.
- **(AI, substantive):** AI-written but factually anchored comments (a sourced summary, a measured comparison with foreign precedent).
- **(human, ragebait):** hand-curated examples of inflammatory but plausibly human comments, with casual register and irregular punctuation.
- **(human, substantive):** ordinary news-comment contributions with specific personal observation or numerical context.

The README documents how to add new test cases and how to interpret outputs in light of the design choices in Section 3.

5 Further development

5.1 Deployment paths

Browser extension. The natural retail deployment is a browser extension that scores comments inline on news portals and major social platforms. Comments are scored client-side (with the lightweight features) or via a privacy-preserving inference gateway that retains no comment text. A visual badge (e.g. small AI-ness and ragebait-ness bars next to each comment) gives the reader a calibrated cue without removing content. Optional aggressive modes can collapse high-scoring comments by default with a click-to-expand affordance.

Editor / newsroom dashboard. News organizations moderating their own comment sections are a natural institutional customer. A dashboard sorts incoming comments by joint risk and surfaces the top red flags, letting one moderator triage what previously required a team.

Platform API. For platforms willing to integrate, the same pipeline can power an upstream filter (rate-limiting or shadow-restricting accounts whose account-level posterior $P(Z_u = 1)$ exceeds a threshold).

5.2 Coordinated inauthentic behavior (CIB) detection

Per-comment scoring detects the leaves; CIB detection finds the trees. We propose, as a population-level extension, the following pipeline:

1. Compute sentence-transformer embeddings $\varphi(x)$ for all comments above an AI-score threshold over a fixed time window.
2. Cluster the embeddings with HDBSCAN. The expected behavior under genuine activity is many small, dispersed clusters; under bot-farm activity, a few unusually tight clusters appear, each corresponding to a single prompt template rotated through many accounts.
3. For each suspect cluster, compute (a) the entropy of the username distribution, (b) the entropy of the posting-time distribution, (c) the cosine-similarity tightness $\bar{S} = \text{mean}_{x \neq y} \cos(\varphi(x), \varphi(y))$. Bot-farm clusters score anomalously high on all three.

This complements per-comment detection at exactly the scale where it is most damaging: a bot farm of 200 accounts each posting one comment per day will not trigger any individual account-level posterior, but the cluster signature is unmistakable.

5.3 Engagement-cascade modeling

A further extension worth flagging: ragebait has a characteristic reply-cascade signature. Modeling the reply process as a Hawkes self-exciting point process [22],

$$\lambda(t) = \mu + \sum_{t_i < t} \alpha e^{-(t-t_i)/\beta},$$

ragebait shows a high α (each reply triggers many follow-ons) but a short β (the cascade burns out fast in hours), while substantive discussion has lower α and longer β (sustained engagement over days). The point-process parameters can be estimated by maximum likelihood and used as a population-level confirmation channel: a comment with high $\hat{p}_{\text{RB}}$ whose subsequent thread fits the Hawkes signature is confirmed; one whose thread does not may be a false positive.

5.4 Continual learning

Each comment that a moderator confirms as AI-generated or as ragebait provides a positive label; each that a moderator clears provides a negative label. A monthly retraining loop, with careful label-noise handling (e.g. inverse-propensity weighting to correct for the fact that we observe more labels for borderline comments than for clear ones), keeps the classifiers current.

The lexicon-based features ($\psi_6, \chi_3, \chi_4, \chi_5$) admit a more direct continual-update strategy: phrases newly-enriched in the AI-positive class can be added to $\mathcal{L}_{\text{AI}}$ on a rolling basis. The ragebait template corpus $\mathcal{R}$ similarly grows with each confirmed positive.

5.5 Closing the template-leakage gap

The first item to address in any continuation of this work is the template-leakage caveat (§3.9). The fix is straightforward: add a `--holdout-frac` flag to `data_collection.py` `templates` that reserves a fraction of labelled-ragebait comments out of $\mathcal{R}$ for evaluation only. Re-running evaluation with the held-out split would yield an honest estimate of the semantic family’s out-of-distribution contribution and would let the rhetorical, neutralization, and info-affect families step into the role the leakage currently masks. We project the corrected Stage 2 AUC at ~ 0.93 – 0.95 depending on holdout fraction.

5.6 Multilingual expansion

The pipeline is largely language-portable. The lexicon-based features require a per-language VAD lexicon, an MFD translation, and a hedge/AI-phrase lexicon. NRC-VAD has translated editions for ~ 100 languages [5]; MFD has community translations for major European languages and a few non-European ones. The remaining features (DetectGPT, GLTR, neutralization, semantic similarity, topic residual) are language-agnostic given a multilingual reference model.

5.7 How long can the detector remain effective?

The AI-side features (likelihood, lexical, punctuation) have a known shelf life: as commercial LLMs improve at imitating short, error-prone, idiomatic comment-section writing, the likelihood gap shrinks. We estimate a 12–24 month useful horizon for these features without active counter-measures.

The ragebait-side features have a much longer horizon. Engineered outrage relies on a small set of psychological levers — moral charge, outgroup framing, intensity saturation, rhetorical confrontation, information vacuum — that are dictated by audience psychology rather than by generation technology. As long as comment platforms reward engagement over substance, the rhetorical structure persists, and the detector with it.

The most durable signal is the counterfactual neutralization gap χ_7 : it is a definition-driven, not pattern-driven, feature. So long as ragebait is operationally defined as “affect with low information,” a paraphraser-based measurement of exactly that quantity remains valid.

6 References

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